University of Toronto Mississauga

Mathematical and Computational Sciences
MAT102H5F - Term Test 2
Test Booklet
Duration - 110 minutes
No Aids Permitted

Surname/Family Name: ____________________

First/Given Name: ____________________

Student Number: ____________________

UTORid: ____________________

Email: ____________________

This exam contains 6 pages (including this cover page) and 5 problems. Check to see if any pages are missing and ensure that all required information at the top of this page has been filled in. You must also have the corresponding Multiple Choice Booklet, which contains the multiple choice questions needed for Question 5 (page 6). No aids are permitted on this examination. Examples of illegal aids include, but are not limited to textbooks, notes, calculators, or any electronic device.

Each part of Question 5 includes five multiple choice answers, of which only a single response is correct. All answers should be indicated on the included scantron sheet (page 6). When filling out the scantron:

- Use a #2 pencil to bubble answers (do not use ink). A #2 pencil ensures that the marks are dark enough.
- Ensure that bubbles are completely filled and make complete erasures.

Any failure to follow these instructions which results in the scanner not being able to read your work will result in a 3 mark deduction. You will not be penalized for incorrect answers.

Your solutions to Problems 1 - 4 must be written in the space provided. No additional paper may be submitted for grading. Do not, under any circumstances, write on the QR code in the top right corner of each page. **Doing so will result in you receiving a zero (0) on this test.**

Unless otherwise indicated, you are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work** in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work, will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit.
- You may use any result mentioned explicitly in the readings, assignments, or in-class worksheets without proof. To use a result which is not covered by the aforementioned sources, you must provide a proof of the result as part of your work.

The Multiple Choice Booklet is printed one-sided, and may thus be used for rough work.

1. (10 points) Show that there are no positive integers x, y such that $x^2 = 8y^2$.

Solution:

Solution 1: We proceed by contradiction. Suppose that x, y are positive integers satisfying $x^2 = 8y^2$. Taking square roots of both sides of the equation, we have $\sqrt{x^2} = \sqrt{8y^2}$ and so $x = 2\sqrt{2}y$. Here we are able to simplify the expression under square root since we know both x and y are positive. Therefore $\sqrt{2} = x/(2y)$, and since both x, y are positive integers we have shown $\sqrt{2}$ is a rational number. This is a contradiction.

Solution 2: We proceed by contradiction. Suppose that there are positive integers satisfying the equation $x^2 = 8y^2$. By the well-ordering principle, we can find positive integers x_0, y_0 with $x_0^2 = 8y_0^2$ such that x_0 is minimal (i.e. for any other positive integers x, y with $x^2 = 8y^2$ we have $x \geq x_0$). Since $x^2 = 8y^2$, x^2 is even and therefore x is even. Therefore $x = 2k$ for some $k \in \mathbb{N}$, and so $8y^2 = 4k^2$, and therefore $2y^2 = k^2$. Since k^2 is again even, we have that k is also even and so $k = 2\ell$ for some $\ell \in \mathbb{N}$. Therefore $2y^2 = 4\ell^2$, and so $y^2 = 2\ell^2$. Since y^2 is even, y is also even and so $y = 2m$ for some $m \in \mathbb{N}$. Finally, plugging $x = 2k, y = 2m$ into the original equation, we have that $4k^2 = 32m^2$ and so $k^2 = 8m^2$. Therefore (k, m) is another positive integer solution to the equation $x^2 = 8y^2$. Since $k = x_0/2$ we have that $k < x_0$, which contradicts the minimality of x_0.

Solution 3: We proceed by contradiction. Suppose that x, y are positive integers satisfying $x^2 = 8y^2$. By the fundamental theorem of arithmetic, there are distinct primes $p_1 = 2, p_2, \ldots, p_k$ and non-negative integers $a_1, \ldots, a_k, b_1, \ldots, b_k$ such that

$$x = 2^{a_1} p_2^{a_2} \cdots p_k^{a_k}, \qquad y = 2^{b_1} p_2^{b_2} \cdots p_k^{b_k}.$$

Plugging these decompositions into the equation $x^2 = 8y^2$ and using that $8 = 2^3$, we have that

$$2^{2a_1} p_2^{2a_2} \cdots p_k^{2a_k} = 2^{3+2b_1} p_2^{2b_2} \cdots p_k^{2b_k}.$$

By the uniqueness of the the prime decomposition in the fundamental theorem of arithmetic, we have that $2a_1 = 3 + 2b_1$, and so $a_1 = b_1 + 3/2$. This contradicts that both a_1, b_1 are integers.

Rubric:

- 1 point for identifying that this is a contradiction proof.
- 6 points for a correct argument, whichever it might be (there are many ways of answering this problem).
- 3 points for clarity of writing.

2. Find *all* of the solutions to the linear Diophantine equation $125x + 90y = 20$.

Solution: Applying the Euclidean algorithm, we get

$$125 = 90 \cdot 1 + 35 \tag{1}$$
$$90 = 35 \cdot 2 + 20 \tag{2}$$
$$35 = 20 \cdot 1 + 15 \tag{3}$$
$$20 = 15 \cdot 1 + 5 \tag{4}$$
$$15 = 5 \cdot 3 + 0 \tag{5}$$

from which we determine that $\gcd(125, 90) = 5$. Since $5 \mid 20$, we know that this linear Diophantine equation has a solution.

Next we solve Bezóut's Identity by reversing the steps above. Starting with (3) we get

$$\begin{aligned} 5 &= 20 + 15(-1) && \\ &= 20 + [35 + 20(-1)](-1) = 35(-1) + 20(2) && \text{from (3)} \\ &= 35(-1) + [90 + 35(-2)](2) = 35(-5) + 90(2) && \text{from (2)} \\ &= [125 + 90(-1)](-5) + 90(2) = 125(-5) + 90(7) && \text{from (1)} \end{aligned}$$

Thus $(x, y) = (-5, 7)$ is a solution to the linear Diophantine equation $125x + 90y = 5$. To get our solution, we multiply everything by 4, yielding $(x, y) = (-20, 28)$.

Finally, the general solutions are given by

$$x = -20 + \frac{90}{5}n = -20 + 18n \quad \text{and} \quad y = 28 - \frac{125}{5}n = 28 - 25n,$$

for $n \in \mathbb{Z}$.

Solution 2: The solutions to the equation $125x + 90y = 20$ are also the solutions to the equation $25x + 18y = 4$, obtained by dividing the first equation by 5. Applying the same steps as above:

$$25 = 18 \cdot 1 + 7 \tag{6}$$
$$18 = 7 \cdot 2 + 4 \tag{7}$$
$$7 = 4 \cdot 1 + 3 \tag{8}$$
$$4 = 3 \cdot 1 + 1 \tag{9}$$
$$3 = 1 \cdot 3 + 0 \tag{10}$$

from which we determine that $\gcd(25, 18) = 1$. Since $1 \mid 4$, we know that this linear Diophantine equation has a solution.

Next we solve Bezóut's Identity by reversing the steps above. Starting with (3) we get

$$\begin{aligned} 1 &= 4 + 3(-1) && \\ &= 4 + [7 + 4(-1)](-1) = 7(-1) + 4(2) && \text{from (3)} \\ &= 7(-1) + [18 + 7(-2)](2) = 7(-5) + 18(2) && \text{from (2)} \\ &= [25 + 18(-1)](-5) + 18(2) = 25(-5) + 18(7) && \text{from (1)} \end{aligned}$$

Thus $(x, y) = (-5, 7)$ is a solution to the linear Diophantine equation $25x + 18y = 1$. To get our solution, we multiply everything by 4, yielding $(x, y) = (-20, 28)$.

Finally, the general solutions are given by

$$x = -20 + \frac{18}{1}n = -20 + 18n \quad \text{and} \quad y = 28 - \frac{25}{1}n = 28 - 25n,$$

for $n \in \mathbb{Z}$.

Rubric:

- 3 points for the Euclidean algorithm.
- 3 points for correctly reversing the Euclidean algorithm to solve Bezout's identity.
- 2 points for the general solutions.
- 2 points for clarity of writing.

3. Consider the relation $\sim$ on $\mathbb{N} \times \mathbb{N}$ given by

$$(a, b) \sim (m, n) \text{ if } a + n = m + b.$$

Determine whether $\sim$ is an equivalence relation.

Solution:

To show that $\sim$ is reflexive, take $(a, b) \in \mathbb{N} \times \mathbb{N}$. Since $a + b = a + b$ holds, we have $(a, b) \sim (a, b)$.

Next, take two elements (a, b) and (m, n) in $\mathbb{N} \times \mathbb{N}$, and assume that $(a, b) \sim (m, n)$. This implies that $a + n = m + b$, which we can write as $m + b = a + n$. Hence $(m, n) \sim (a, b)$ and $\sim$ is symmetric.

Now take three elements $(a, b), (m, n), (r, s) \in \mathbb{N} \times \mathbb{N}$. Assume that $(a, b) \sim (m, n)$ and $(m, n) \sim (r, s)$. From these we get the two equations

$$\begin{aligned} a + n &= m + b \\ m + s &= r + n \end{aligned}$$

Adding both equations together and then subtracting $(m + n)$ from both sides of the result, we get

$$\begin{aligned} a + n + m + s &= m + b + r + n \\ a + s &= r + b \end{aligned}$$

which implies that $(a, b) \sim (r, s)$. This proves transitivity.

Therefore $\sim$ is an equivalence relation.

Note: One could also rewrite the condition for $\sim$ as $a - b = m - n$ which simplifies the proof.

Rubric:

- 2 points for reflexive.
- 2 points for symmetric.
- 4 points for transitive.
- 2 points for clarity of writing.

4. Suppose that p is a prime number.

(i) (8 points) Show that $x^2 \equiv y^2 \pmod{p}$ if and only if $x \equiv y \pmod{p}$ or $x \equiv -y \pmod{p}$.

Solution: We first prove the forward direction. Say $x^2 \equiv y^2 \pmod{p}$. Then $x^2 - y^2 \equiv 0 \pmod{p}$. Now $x^2 - y^2 = (x-y)(x+y)$ tells us that $(x-y)(x+y) \equiv 0 \pmod{p}$. This means $p \mid (x+y)$ or $p \mid (x-y)$. This is the same as $x \equiv \pm y \pmod{p}$.

If $x \equiv y \pmod{p}$ then we know that

$$x \cdot x \equiv y \cdot y \pmod{p}$$

which gives the desired result. Similarly, if $x \equiv -y \pmod{p}$ then

$$x \cdot x \equiv (-y) \cdot (-y) \pmod{p}$$

which gives the same outcome.

Rubric:

- 4 points for the ($\Longrightarrow$) direction.
- 2 points for the ($\Longleftarrow$) direction.
- 2 points for the clarity of writing.

(ii) (2 points) By means of an explicit counter-example, show that this is not true when p is not prime.

Solution: Take $x = 2$ and $y = 4$, and $p = 12$. Then $x^2 \equiv y^2 \pmod{12}$, but clearly $2 \not\equiv \pm 4 \pmod{12}$.

Rubric:

- 1 point for a correct counter example.
- 1 point for showing that the counter-example is in fact a counter-example.